

Outcomes After Operative and Nonoperative Management of Hamstring Injuries

A Systematic Review

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Background: Hamstring injuries are common in sports involving rapid acceleration and directional changes. Despite extensive research, management remains primarily dependent on clinical judgment, owing to the absence of standardized, evidence-based protocols.

Purpose: To compare the outcomes of operative versus nonoperative treatments, focusing on patient satisfaction, muscle strength, range of motion (ROM), activity level, and return to sport.

Study Design: Systematic review; Level of evidence, 4.

Methods: Using PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, the authors performed a systematic search of the PubMed, Google Scholar, Web of Science, and ScienceDigest databases from January 2000 to May 2024. Reported outcomes measured were muscle strength, ROM, functional activity, pain levels, return to preinjury activity levels, and patient satisfaction. Additionally, any complications related to the intervention were documented. The authors identified 13 eligible studies involving adults with acute hamstring injuries, including 526 patients. Eleven studies specifically investigated operative treatment, 1 examined nonoperative management, and 1 directly compared both approaches.

Result: Surgical intervention was consistently associated with superior outcomes across multiple domains. Postoperatively, the Lower Extremity Functional Scale score often exceeded 74, and strength recovery commonly approached 90.0% of the contralateral limb. ROM was preserved after surgery but was not evaluated in nonoperative cohorts. Functional activity was generally higher after operative treatment (mean Tegner Activity Scale [TAS] scores: 4.7 ± 0.8 , 6.0 ± 1.47 , 5.1, and 8.5 ± 2.4 ; Marx Activity Rating Scale scores: 3.5 ± 4.3 and 4.4 ± 4.4), whereas nonoperative treatment showed a decline in TAS score (6.9 ± 1.7 to 6.1 ± 1.9 ; $P = .030$). Pain levels were low overall (visual analog scale score: 0.7 ± 0.9 to 4.0 ± 4.0). Complications were more frequent after surgery (4.2%-36.6%, including hematomas, neurapraxia, and superficial infections), although mostly minor. Nonoperative complications were limited to ecchymosis.

Conclusion: This study demonstrated that operative treatment was associated with good functional recovery, muscle strength, and high return-to-sport rates. Complications rates within the operative group were mostly minor and not statistically significant when compared to nonoperative management, which yielded good outcomes in the limited available data. Further prospective studies using standardized outcome measures and clear subtype definitions are required to determine which patients benefit most from operative versus nonoperative management.

Keywords: muscle injuries; hip/pelvis/thigh; medical aspects of sports; NSAIDs

biceps femoris, which arises from the femur. Functionally, the hamstrings facilitate knee flexion and hip extension, with the biceps femoris contributing to external rotation, while the semimembranosus and semitendinosus support internal rotation.²⁴ Their biarticular characteristics make them susceptible to injuries, particularly the biceps femoris, due to its dual innervation.²⁶ Key risk factors for recurrence include a previous hamstring injury, advanced age, and elevated muscle torque.³⁴ Moreover, ethnicity and sport-specific factors, such as participation in high-level competition and engagement in sports that require frequent high-intensity accelerations, have been shown to increase injury susceptibility.^{13,14}

Soft tissue injuries are classified into 3 grades based on clinical severity. Grade 1 involves minimal structural disruption with mild pain and tenderness, typically elicited only during activity. Grade 2 represents partial tissue disruption with moderate to severe pain and functional limitation, while grade 3 indicates complete rupture or avulsion with marked functional loss; paradoxically, pain may be absent due to tissue discontinuity. Although grading is primarily clinical, magnetic resonance imaging (MRI) and ultrasound are often used to confirm the diagnosis, particularly in grades 2 and 3.²⁰ This system is similarly applied to hamstring injuries.¹⁶

Management is largely guided by clinical judgment, as no universally accepted evidence-based protocol exists. Grades 1 and 2 are generally treated nonoperatively with rest, ice, short-term nonsteroidal anti-inflammatory drugs, and structured rehabilitation. The use of platelet-rich plasma and other biological therapies remains controversial given limited supporting evidence and potential World Anti-Doping Agency restrictions related to insulin-like growth factor-1.^{4,11} Initial care emphasizes hemorrhage, edema, and pain control, followed by assisted ambulation and postural correction.²³

In contrast, grade 3 injuries often necessitate surgical intervention, particularly when tendon retraction exceeds 2.0 cm, as nonoperative approaches may result in persistent pain, weakness, and long-term functional deficits. Early surgical repair—ideally within 4 to 6 weeks—is recommended to reduce complications such as sciatic nerve scarring.¹⁰ Adjunct modalities, including therapeutic ultrasound and shockwave therapy, have demonstrated limited or no significant clinical benefit in this context.²⁷

This systematic review evaluates and compares operative and nonoperative treatments for acute hamstring injuries by analyzing outcomes related to muscle strength,

range of motion (ROM), functional activity, pain levels, return to preinjury activity, patient satisfaction, and treatment-related complications. We hypothesized that operative and nonoperative treatments for acute hamstring injuries would demonstrate differences in patient satisfaction, functional recovery (hamstring ROM, muscle strength, and functional activities), return to preinjury activity levels, and complication rates.

METHODS

This systematic review was conducted from January 2000 to May 2024 in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines.²⁵ All stages of the review process followed the methodological framework outlined in the *Cochrane Handbook for Systematic Reviews of Interventions*.¹⁷ The study protocol was prospectively registered in PROSPERO (International Prospective Register of Systematic Reviews) under the identification number CRD42024567841 (<https://www.crd.york.ac.uk/PROSPERO/view/CRD42024567841>).

Eligibility Criteria

Studies were included if they met the following criteria:

- (1) Included patients aged ≥ 18 years with acute hamstring muscle injuries, classified by anatomic location as ischial tuberosity avulsions, intramuscular tears, or musculotendinous junction injuries, treated either operatively or nonoperatively;
- (2) Reported treatment-related outcomes, including patient satisfaction, functional outcomes (hamstring ROM, muscle strength, and functional activity), and return to preinjury activity levels;
- (3) Included interventions initiated within 6 weeks of injury, with the acute nature of the injury clearly stated^{6,28};
- (4) Published in the English language;
- (5) Were a randomized controlled trial, observational study, or case series with a minimum of 10 patients.

Exclusion criteria were as follows: studies that did not report outcomes of interest; injuries classified as chronic (ie, treated >6 weeks after onset) or where acute versus chronic injury was not clearly distinguished; non-hamstring-specific injuries; and articles published in languages other than English or outside the specified time frame.

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Ethical approval was not required, as this study is a systematic review of previously published studies and did not involve new data collection from human or animal subjects.

Search Strategy

A comprehensive literature search was conducted in the PubMed, Google Scholar, Web of Science, and ScienceDirect databases. The search strategy used Boolean operators and relevant keywords, including “Hamstring muscle injury,” “Operative treatment,” and “Conservative treatment” (Appendix 1).

The Rayyan web-based tool (www.rayyan.ai), which utilizes artificial intelligence for screening and deduplication, was used to manage the selection process. Each step was independently verified manually by 2 reviewers (D.O.S. and A.A.Alshammmary) to ensure accuracy and minimize selection bias.

The initial search yielded 1205 articles. After duplicate removal via Rayyan and manual verification, 815 unique articles remained for screening. A 2-stage screening process was conducted: titles and abstracts were screened first, followed by full-text review. Discrepancies were resolved by a third reviewer (A.A.Alharbi) in a blinded manner. Studies that fulfilled the predefined inclusion criteria were selected for final analysis.

Data Extraction

Data were extracted from full-text articles and systematically recorded in a Microsoft Excel spreadsheet. Each study was analyzed across several domains.

Study characteristics included article title, author names, country of origin, study design, sample size, participant age, sex distribution, body mass index, inclusion criteria, and group categorization (if applicable).

Injury-related variables included anatomic site, operational definition of acute injury, intervention protocol, follow-up duration, and time from injury to surgery. Reported outcomes included muscle strength, ROM, pain, return to preinjury activity, patient satisfaction, and functional activity. Functional recovery was evaluated using validated instruments, including the Lysholm score, passive straight-leg raise (PSLR), modified Harris Hip Score (HHS), Marx Activity Rating Scale (MARS) score, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), straight-leg raise test score, limb symmetry index (LSI) for flexor strength, Lower Extremity Functional Scale (LEFS) score, Perth Hamstring Assessment Tool (PHAT) score, International Hip Outcome Tool-12 (iHOT-12) score, and Kerlan-Jobe Orthopaedic Clinic (KJOC) score. Functional activity after both operative and nonoperative treatments was assessed using the Short Form-12 (SF-12) score, Tegner Activity Scale (TAS) score, and MARS score. Pain levels were consistently assessed using the visual analog scale (VAS) score.

Complications related to the interventions were also recorded. All data were independently reviewed by 2 authors (R.H.A. and O.H.A.), and any discrepancies during data extraction were resolved by a third reviewer (S.D.A.) through blinded adjudication to ensure objectivity and minimize bias.

Quality Assessment

The methodological quality of the included studies was independently assessed by 2 reviewers (R.H.A. and D.O.S.), with a third reviewer (K.M.A.) performing a final evaluation (Figure 1) and resolving any discrepancies. For nonrandomized cohort studies, the Newcastle-Ottawa Scale was used, categorizing studies into predefined quality levels: high quality (7-9 stars), moderate quality (4-6 stars), and low quality (0-3 stars).³² In case series, a validated checklist was applied to classify studies as high quality (8-10 “yes” responses), moderate quality (5-7 “yes” responses), and low quality (0-4 “yes” responses).⁷

RESULTS

Characteristics of Included Studies

A total of 13 articles[#] were included, addressing various operative and nonoperative interventions for acute hamstring ruptures, whether partial or complete. A summary of study characteristics is provided in Table 1, while the baseline features are detailed in Table 2.

Results Synthesis

Within the included studies, 11 investigated surgical repairs,^{**} 1 focused exclusively on nonoperative management,¹² and 1 study compared both surgical and nonoperative treatments³⁰ (Table 3).

Functional Recovery

Operative Treatment. Across the included studies, 458 patients were analyzed. One study reported a mean postoperative LEFS score of 74.9 ± 7.8 (range, 59-80) and HHS of 90.7 ± 13.9 (range, 67-100).⁸ Another study reported a mean Lysholm score of 90.2 ± 3.4 , a mean WOMAC score of 96.3 ± 2.4 , and a mean University of California, Los Angeles (UCLA) activity score of 8.1 ± 0.9 .²⁹ The iHOT-12 (mean, 99) and KJOC (mean, 77) scores were also reported, with 78.0% of patients demonstrating good or excellent outcomes.⁵ Additional reports described a mean Lysholm score of 85.7 ± 18.3 and a mean modified HHS of 91.0 ± 12.6 .³³ Functional recovery over time was assessed using the PSLR, MARS score, and LEFS score, all of which showed significant improvement by the 3-month follow-up.¹ At 2 years, 30.0% of patients achieved LEFS scores of 80.0, and 45.0% had MARS scores of 75.0.¹ Among 64 surgically treated musculotendinous junction injuries of the long head of the biceps femoris, LEFS and MARS scores improved significantly at 3 months, 1 year, and 2 years.² LEFS scores of 80.3 ± 20.2 and 94.3 ± 10.1 were observed in 2 postoperative groups.¹⁹ The

[#]References 1, 2, 5, 8, 9, 12, 19, 21, 22, 29-31, 33.

^{**}References 1, 2, 5, 8, 9, 19, 21, 22, 29, 31, 33.

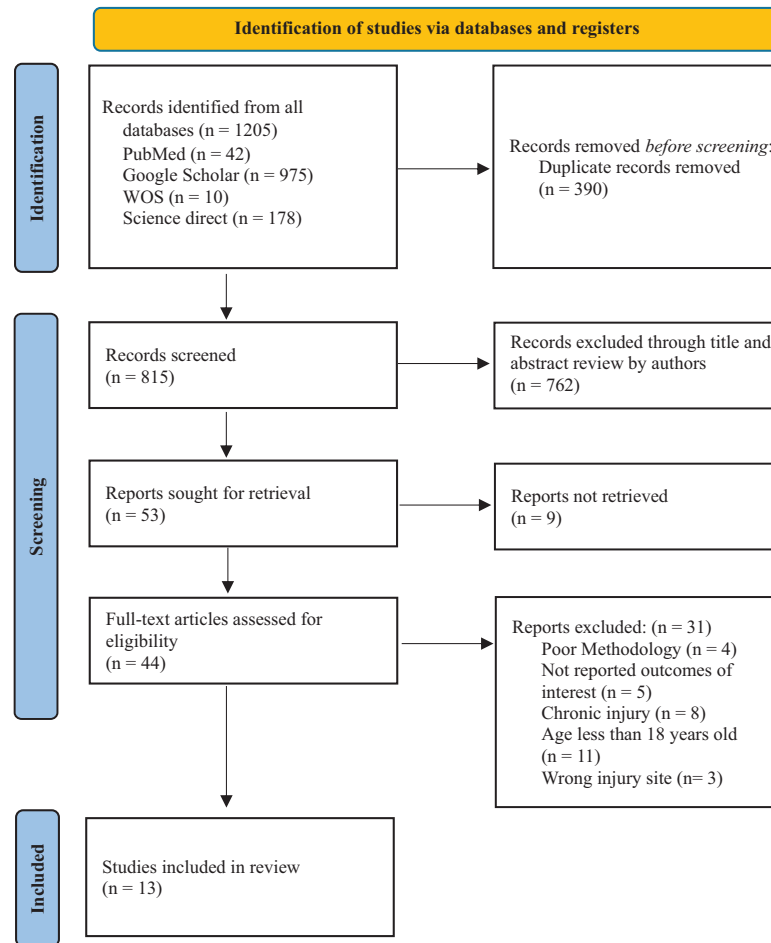


Figure 1. PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flowchart of included articles. WOS, Web of Science.

median EQ-5D-5L score was 0.94 (0.83-1.00), with an LEFS score of 88.8 (range, 61.6-100.0), PHAT score of 78.8 (range, 54.6-99.8), and LSI score of 95.6% (range, 71.7%-135.6%).⁹

Nonoperative Treatment. Only 1 study assessed nonoperative management for unilateral proximal hamstring rupture, including 16 patients. It reported a mean LEFS score of 76.6 ± 3.9 and a UCLA activity score of 37.5 ± 2.2 .¹²

Operative Versus Nonoperative Treatment. In the study by Shambaugh et al,³⁰ comparisons between the operative group (n = 14) and the nonoperative group (n = 11) demonstrated higher mean LEFS scores in the operative group (74.71 ± 5.38 vs 68.50 ± 7.92). Additionally, the mean single-leg hop distance was significantly greater in the operative group (119.1 ± 27.7 cm) than in the nonoperative group (56.1 ± 31.2 cm).³⁰

Muscle Strength

Operative Treatment. In 1 study, a mean postoperative hamstring strength recovery of $78.0\% \pm 6.1\%$ (range,

74.0%-88.0%) was observed compared with the contralateral side.⁸ In another study, strength restoration $>90.0\%$ was observed at the 1-year follow-up,¹ with significant improvement from preoperative to postoperative levels confirmed.² Mean muscle strength values of 80.6 ± 20.1 and 91.8 ± 10.6 were reported in 2 operative cohorts.¹⁹ In the study by Chocholáč et al,⁹ the median maximum flexor strength was 0.82 Nm/kg (range, 0.33-1.42 Nm/kg) on the operated side compared with 0.72 Nm/kg (range, 0.41-1.89 Nm/kg) on the contralateral side ($P = .807$), with a median LSI of 95.6% (range, 71.7%-135.6%).

Nonoperative Treatment. In 1 study, a concentric strength deficit of $40\% \pm 25.0\%$ was observed at 4 months, decreasing to $25.0\% \pm 12\%$ at 2 years.¹² However, the eccentric strength deficit persisted over time.

Operative Versus Nonoperative Treatment. In the study by Shambaugh et al,³⁰ the mean perceived strength was $86.07\% \pm 12.12\%$ in the operative group compared with $83.64\% \pm 14.16\%$ in the nonoperative group.³⁰ The mean isokinetic strength in the operative group was $90.87\% \pm 16.3\%$. In contrast, isometric testing in the nonoperative group revealed marked weakness at 45° ($57.54\% \pm 7.8\%$) and 90° ($67.73\% \pm 18.8\%$) of flexion compared to the

TABLE 1
Summary Characteristics of Included Studies^a

Study	Country	Study Design	Sample Size	Inclusion Criteria	Intervention	Primary Outcome	Main Conclusion
Shambaugh ³⁰	USA	Cohort study	25 patients	≥12 months follow-up from surgery or injury.	Nonoperative and operative treatment of complete proximal hamstring ruptures.	LEFS scores: 68.50 ± 7.92 (nonoperative) vs 74.71 ± 5.38 (operative) ($P = .07$). No difference in SF-12 or single-leg hop distance. Nonoperative group showed weakness at 45° (57.54% ± 7.8%) and 90° (67.73% ± 18.8%). Operative group showed 90.87% ± 16.3% strength. All operative patients returned to preinjury activities vs 3 nonoperative patients unable to return (chi-square = 4.33; $P = .07$).	Acute surgical treatment restored ~90% strength and increased likelihood of return to preinjury activities compared to nonoperative care.
Willinger ³³	Germany	Case series	94 patients	Confirmed proximal hamstring injury by clinical exam + preoperative MRI; partial (<3 ruptured tendons) or complete tear.	Proximal hamstring tendon repair.	Clinical outcomes were excellent with no difference between groups or injury patterns. RTS 86.2%, significantly higher after acute treatment ($P < .05$). Complication rate 8.5%, higher in complete tears and delayed surgery ($P < .05$).	Surgical treatment yields excellent outcomes and high RTS. Early surgery provides higher RTS and fewer complications than delayed repair. Partial and complete tears show similar outcomes, but complete avulsions have more complications.
Sandmann ²⁹	Germany	Case series	15 patients	Patients with 16 proximal hamstring ruptures.	Surgical treatment by osseous repair of the hamstring tendons using suture anchor systems (Mitek Super Anchors, Mitek; Titanium Cork Screw, Arthrex).	Lysholm, TAS, UCLA, and WOMAC scores showed return to preinjury level. Operated leg strength similar to uninjured ($P > .094$). No significant differences in sport type/frequency ($P > .05$). Surgical repair gives good midterm results with similar isokinetic strength to healthy leg.	Most surgically treated patients returned to sport at preinjury level at midterm follow-up. Consistently good to excellent scores, with similar flexibility and strength to uninjured leg. Surgical refixation recommended for active patients; low complication rate.
Chocholáč ⁹	Switzerland	Case series	13 patients	Age >18 y; surgically treated proximal hamstring ruptures with modified anchor placement (2014-2019).	Surgically treated for acute proximal hamstring rupture using modified anchor placement.	Median PHAT score 78.8 (range, 54.6-99.8), EQ-5D-5L score 0.94 (range, 0.83-1.00), LEFS score 88.75 (range, 61.25-100). Median satisfaction 100% (range, 90%-100%). One patient reported ischial tuberosity discomfort. No difference in hip flexion or isokinetic strength ($P > .58$). Clinical scores did not correlate with LSI ($P > .125$). No reruptures or sciatic radiculopathy.	Modified extra-anatomic anchor placement yielded good clinical and functional outcomes, no pain while sitting, and muscle strength comparable to contralateral side.

(continued)

TABLE 1
(continued)

Study	Country	Study Design	Sample Size	Inclusion Criteria	Intervention	Primary Outcome	Main Conclusion
Chahal ⁸	USA	Case series	15 patients	Diagnosis confirmed by history, clinical exam, and MRI; at least 1 of acute posterior thigh pain, audible "pop," or tearing sensation at injury.	Surgical repair.	13/15 patients followed for mean of 36.9 mo. LEFS score 74.9 ± 7.8 , HHS 90.7 ± 13.9 , SANE score 93.6 ± 7.5 , VAS score 1.3 ± 1.9 , TAS score 4.6 ± 2.3 . All athletes returned to sport, 45% at reduced level. Isokinetic strength recovered to $78.0\% \pm 6.1\%$. MRI showed 100% healed repair; 3/12 had tendinopathy/mild atrophy.	Surgical repair of complete hamstring ruptures provides pain relief, good outcomes, high satisfaction, and excellent healing, but does not fully restore preinjury function or sport level.
Fouasson-Chailloux ¹²	France	Case series	16 patients	Complete proximal rupture of ≥ 1 tendon, age >18 y, injury during sport; excluded: distal rupture, bone avulsion, or surgical management.	Nonoperative therapy.	All patients returned to sport at 7.0 ± 2.9 mo, same or lower level. TAS score decreased from 6.9 ± 1.7 to 6.1 ± 1.9 ($P = .03$). 15 patients were satisfied. Concentric strength deficit improved from $40\% \pm 25\%$ at 4 mo to $25\% \pm 12\%$ at 2 y; eccentric deficit persisted.	Subjective results were good despite persistent isokinetic strength deficit.
Ayuob ¹	UK	Case series	20 patients	Injury within 4 wk; MRI-confirmed complete nonavulsion BAMIC grade 4 tear of proximal semimembranosus tendon; surgery by senior author.	20 professional athletes undergoing acute primary surgical repair of complete nonavulsion proximal semimembranosus injuries confirmed on preoperative MRI.	19/20 (95%) patients returned to preinjury sport. RTS time 11.9 ± 5.7 wk. No recurrences. At 3 mo, significant improvement in PSLR, isometric strength (0° , 15° , 45° ; all $P < .001$), LEFS score (64.8 ± 4.6 vs 34.4 ± 5.1), and MARS score (10.7 ± 1.6 vs 5.5 ± 2.0). High satisfaction maintained at 1 and 2 y.	Acute repair of complete proximal semimembranosus injuries produced high satisfaction, improved strength, functional scores, and high RTS with low recurrence at short-term follow-up.
Ayuob ²	UK	Case series	64 patients	Acute hamstring injury within 4 wk; MRI-confirmed BAMIC grade 3B or 4 tear of proximal MTJ-BFh; clinical loss of strength/flexibility; surgery by senior author.	Surgical repair of acute MTJ-BFh injuries.	3 reinjuries (1.6% MTJ-BFh, 3.2% myofascial). At 3 mo, PSLR, isometric strength (0° , 15° , 45°), LEFS score, and MARS score all significantly improved ($P < .001$). High satisfaction and functional outcomes maintained at 1 and 2 y.	Surgical repair of acute MTJ-BFh injuries allows RTS at preinjury level with low recurrence risk at short-term follow-up.
Mansour ²²	USA	Case series	10 patients	NFL players with unilateral proximal hamstring rupture (1990-2008); data collected via physician questionnaire (demographics, injury details, management, RTP, games missed/played).	Surgical intervention.	10 injuries surgically fixed within 10 days. All regained strength; 9/10 returned to play. Only 5/10 played >1 game despite no surgical limitations.	Outcomes are better for surgically repaired acute proximal hamstring ruptures than neglected ones. 90% of NFL players returned to same level, but only 50% sustained RTP. Trend toward significance for draft status ($P = .11$).

(continued)

TABLE 1
(continued)

Study	Country	Study Design	Sample Size	Inclusion Criteria	Intervention	Primary Outcome	Main Conclusion
Johnson ¹⁹	USA	Cohort study	20 patients	Patients with WC claim at time of proximal hamstring avulsion repair + ≥ 2 -y follow-up; complete tear of all 3 tendons; control group matched by age, sex, BMI.	Open repair of complete proximal hamstring avulsions.	WC group (n = 20) had lower LEFS score (69.1 vs 94.1; $P < .001$), lower return-to-work rate (70% vs 94.1%; $P = .039$), and longer RTW time (4.3 vs 3.5 months; $P = .029$) compared with controls. No differences in age, sex, BMI.	WC patients had inferior outcomes, slower RTW, and lower RTW rate than controls.
Bowman ⁵	USA	Case series	58 patients	Patients with complete or partial proximal hamstring avulsion + ≥ 12 -mo follow-up; excluded: skeletally immature, avulsion fracture, prior repair, allograft reconstruction, other extremity injuries/surgery.	Proximal hamstring repairs.	Of 102 repairs, 86 were eligible, 58 were enrolled and analyzed (67%). Satisfaction 94%. RTS 88% at 7.6 mo; 72% at same level. Runners returned at 6.3 mo, 50% at same level, fewer miles/wk (15.7 vs 7.8; $P < .001$). 78% had good/excellent SANE scores. TAS score decreased (5.5 to 5.1). iHOT-12 score 99, KJOC score 77. Open vs endoscopic repairs had similar outcomes. Greater satisfaction noted in patients > 50 y ($P = .024$) but less likely to return to running ($P = .010$).	Satisfaction and functionality high. No significant outcome differences by demographics, BMI, smoking, tear grade, or technique. Acute tears had better SANE scores. Runners may not return to preinjury level.
Lefèvre ²¹	France	Case series	64 patients	Professional athletes (remunerated, national/international competition level) with surgically treated PHCI; excluded: revision surgery, < 1 -y follow-up, bony PHAI, refusal to enroll.	Surgical repair of proximal hamstring complex injuries.	64 athletes (mean age, 27.3 y) had RTS 98.4%, with 78.1% at preinjury level by 6.2 mo. Male sex, higher TAS score, semimembranosus or conjoint tendon injury, and PHTR associated with better RTS (all $P < .05$). Complication rate 4.7%.	Favorable predictors of early RTS: male sex, isolated semimembranosus injury, PHTR injuries.
Subbu ³¹	UK	Case series	112 patients	Operatively managed complete avulsion injuries with ≥ 2 -y follow-up, comparing timing of surgery and return to preinjury sport.	Surgery for complete proximal hamstring.	RTS achieved in 96.4%. Early intervention group returned at 16 wk, 9 wk faster than delayed group. 2 partial reruptures occurred in delayed group. 12 athletes delayed by nerve symptoms; 2 required explorations.	Early surgical intervention leads to faster RTS and fewer complications; delayed diagnosis increases morbidity.

^aBAMIC, British Athletics Muscle Injury Classification; BMI, body mass index; HHS, Harris Hip Score; iHOT-12, International Hip Outcome Tool-12; KJOC, Kerlan-Jobe Orthopaedic Clinic; LEFS, Lower Extremity Functional Scale; LSI, limb symmetry index; MARS, Marx Activity Rating Scale; MTJ-BFlh, musculotendinous junction of the long head of the biceps femoris; MRI, magnetic resonance imaging; NFL, National Football League; PHAI, proximal hamstring tendon avulsion injury; PHAT, Perth Hamstring Assessment Tool; PSLR, passive straight-leg raise; PHCI, proximal hamstring complex injury; PHTR, proximal hamstring free tendon rupture; RTP, return to play; RTS, return to sport; RTW, return to work; SF-12, Short Form-12; SANE, Single Assessment Numeric Evaluation; TAS, Tegner Activity Scale; UCLA, University of California, Los Angeles; VAS, visual analog scale; WC, workers' compensation; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index; wk, weeks; mo, months; y, years.

TABLE 2
Baseline Features of the Included Studies^a

Study	Groups	No. of Participants	Age, y ^b	Male Sex, n (%)	Follow-up ^c	BMI, kg/m ^{2c}	Time From Injury to Surgery, days ^c	Site of Injury	Management
Shambaugh ³⁰	Operative	14	46.98 ± 9.73	5 (35.7)	3.56 ± 2.11 y	NR	28.14 ± 14.06	Complete proximal hamstring ruptures.	Surgical repair of the hamstring.
	Nonoperative	11	55.68 ± 10.45	9 (81.8)	2.48 ± 3.66 y	NR	4.63 ± 3.66		Nonsurgical care consisted of rest, ice, physical therapy, selective corticosteroid injections, and gradual return to activities (~4 mo). All patients followed the same rehabilitation protocol: early mobilization, gentle ROM and flexibility exercises, then strengthening and return to sport within 6-12 wk.
Willinger ³³	Classified by treatment timing and injury pattern	94	53.8 ± 12.3	53 (56.4)	56.2 ± 27.2 mo	27.2 ± 14.2	54.2 ± 117.7	Partial vs complete tears (1 or 2 tendons vs 3 tendons).	Tendons sutured with Krackow stitches and reduced. Sutures tied over the tendon. Postoperatively, the operated leg was braced (hip flexion limited to 30°, extension to 0°), weightbearing restricted to 20 kg for 6 wk. Physical therapy started day 1 with passive flexion and abduction.
Sandmann ²⁹	NR	15	47 (21-66)	9 (60.0)	56 mo (24-112 mo)	NR	64 ± 73	Proximal hamstring complex.	19 ruptures were treated by osseous repair of the hamstring tendons using suture anchor systems (Mitek Super Anchors and Titanium Cork Screw).
Chocholáč ⁹	NR	13	64.2 (52.1-80.4)	5 (38.5)	46.2 mo (11.2-75.0 mo)	28.5 (23.5-45)	Median 14	Right 7, left 6; 12 had total rupture (ST + BFlh conjoint + SM), 1 had partial rupture (ST + BFlh conjoint).	Modified anchor placement was used for acute proximal hamstring rupture.
Chahal ⁸	NR	13	44.6 (26-58)	8 (61.5)	36.9 mo (27-63 mo)	NR	134.8 days (9 days-4 y)	Complete rupture of ≥1 proximal hamstring tendon.	Patients were positioned prone, with a transverse 5- to 8-cm gluteal fold incision. Care was taken to avoid injury to the posterior femoral cutaneous nerve. The gluteus maximus was elevated, fascia opened, and hematoma evacuated in acute cases.

(continued)

TABLE 2
(continued)

Study	Groups	No. of Participants	Age, y ^b	Male Sex, n (%)	Follow-up ^c	BMI, kg/m ^{2c}	Time From Injury to Surgery, days ^c	Site of Injury	Management
Fouasson-Chailloux ¹²	NR	16	34 ± 12 (18-49)	10 (62.5)	73 ± 39 mo (24-144 mo)	NR	NR	Nonoperative.	
Ayuob ¹	NR	20	28.8 ± 4.8	11 (55)	27.6 mo (24.0-34.6 mo)	25.3 ± 3.5	20.9 ± 4.8	Right 14 (70%), left 6 (30%); complete nonavulsion proximal SM injuries.	Acute surgical repair for complete nonavulsion proximal SM injuries.
Ayuob ²	NR	64	26.6 ± 3.9	42 (65.6)	29.2 mo (24.0-37.1 mo)	26.1 ± 2.4	12.4 ± 6.2	Right 37 (57.8%), left 27 (42.2%).	Surgical repair of acute MTJ-BF _{lh} injuries.
Mansour ²²	NR	10	27.2 (23-30)	NR	NR	NR	7 days (3-10 days)	Complete proximal hamstring ruptures.	Surgical Reattachment of complete proximal hamstring ruptures.
Johnson ¹⁹	Workers' compensation patients	15	52.3 ± 7.0		6.1 ± 2.3	32.4 ± 6.8	NR	Proximal hamstring avulsion repair.	Surgical proximal hamstring avulsion repair.
Bowman ⁵	Control	14	50.6 ± 10.3		5.3 ± 2.5	31.2 ± 8.2	NR		
	NR	58	51.1 ± 12.0	25 (43.1)	29.0 ± 9.9	25.3 ± 4.4	176 ± 356	Proximal hamstring avulsions (complete/partial); right 33 (57%).	Surgical technique: open 52 (90%), endoscopic 6 (10%).
Lefevre ²¹	NR	64	27.3 ± 8.9	53 (82.8)	Duration for subsequent follow-ups 4.2 ± 3.9 y	24.5 ± 3.8	Time delay to RTS 6.2 ± 2.5 mo	Hamstring injury side: right 27 (42.2%), left 37 (57.8%).	Surgical treatment for PHCI at a sport surgery center.
Subbu ³¹	Early intervention (<6 wk)	78	29.7 (18-52)	51 (65.4)	Minimum 2 y	NR	22 days (range, 5-42 days)	Complete proximal hamstring avulsion injuries.	Surgical exploration and repair of complete avulsions of the proximal hamstring complex.

^aBMI, body mass index; MTJ-BF_{lh}, musculotendinous junction of the long head of the biceps femoris; NR, not reported; PHCI, proximal hamstring complex injury; ROM, range of motion; SM, semimembranosus; ST, semitendinosus.

^bPresented as mean ± SD or median (range).

^cPresented as mean ± SD or mean (range) unless otherwise indicated.

uninjured side.³⁰ The mean follow-up duration was 3.56 ± 2.11 years for the operative group and 2.48 ± 3.66 years for the nonoperative group.³⁰

Range of Motion

No studies assessed the ROM after nonoperative treatment; however, 4 studies evaluated ROM after surgery. Biarticular and monoarticular hip flexibility showed mean values of $86.6^\circ \pm 8.6^\circ$ and $127.8^\circ \pm 12^\circ$, respectively, in the work of Sandmann et al.²⁹ In the study by Chocholáč et al,⁹ median passive hip flexion with the knee bent was 120° (range, 115° - 140°) on the operated side and 120° (115° - 150°) on the contralateral side ($P = .581$). With the knee straight, median passive hip flexion was 90° (range, 60° - 120°) on the operated side and 90° (70° - 110°) on the contralateral side ($P > .999$).⁹

Functional Activities

Operative Treatment. Postoperative TAS scores were reported as 4.7 ± 0.8 ,²⁹ 6.0 ± 1.47 ,³³ and 5.1 .⁵ Higher

mean TAS scores of 8.5 ± 2.4 ²¹ and 8.5 ± 2.4 ¹⁹ were also documented in other studies.

Nonoperative Treatment. A significant decline in TAS score from 6.9 ± 1.7 to 6.1 ± 1.9 was observed after nonoperative treatment ($P = .030$).¹²

Operative Versus Nonoperative Treatment. No significant differences were found between groups in SF-12 mental and physical component scores, indicating comparable functional outcomes.³⁰

Patient Satisfaction

Operative Treatment. All patients were reported to be extremely satisfied postoperatively.⁸ High satisfaction scores of 93.1 ± 13.1 and 96.7 ± 5.3 were documented in 1 study.¹⁹ Median satisfaction rates were 94.0%,⁵ 95.0%,¹ and 100.0%.⁹ Although 2 patients were unsatisfied at the 1-year follow-up in the study by Ayuob et al,² satisfaction improved at 2 years, with 81.2% reporting "very satisfied" and 18.8% reporting "satisfied."

Nonoperative Treatment. Fifteen of 16 patients were either very satisfied or satisfied, with high subjective outcome scores.¹²

TABLE 3
Outcomes of the Included Studies^a

Study	Groups	Functional Recovery	Muscle Strength	ROM	Functional Activities	Patient Satisfaction	Return to Preinjury Activity Levels	Pain Levels	Adverse Effects or Complications
Shambaugh ³⁰	Operative	Mean LEFS score 74.71 ± 5.38; single-leg hop distance 119.1 ± 27.7 cm.	Perceived strength 86.07% ± 12.12%; isokinetic testing showed 90.87% ± 16.3% strength compared to the uninjured leg.	NR	SF-12 mental and physical component scores were similar between groups.	NR	All patients in operative group returned to preinjury activities.	NR	1 reoperation for hematoma; 1 case of incisional numbness, resolved after 3 mo.
	Nonoperative	Mean LEFS score 68.50 ± 7.92; single-leg hop distance 56.1 ± 31.2 cm.	Perceived strength 83.64% ± 14.16%; isometric testing showed significant weakness at 45° (57.54% ± 7.8%) and 90° (67.73% ± 18.8%) of flexion.	NR		NR	3/11 patients in nonoperative group did not return to preinjury activities.	NR	NR
Willinger ³³	Classified by treatment timing and injury pattern	Mean Lysholm score 85.7 ± 18.3; modified HHS 91.0 ± 12.6.	NR	NR	Mean TAS score 6 (range, 2-9).	NR	RTS: same level 36 (50.0%), reduced 30 (41.7%), stopped 6 (8.3%).	VAS score 1.3 ± 1.8	3 (4.2%) postoperative complications; sitting discomfort in 26 (36.1%).
Sandmann ²⁹	NR	Mean ± SD Lysholm score 90.2 ± 3.4; WOMAC score 96.3 ± 2.4; UCLA score 8.1 ± 0.9.	NR	Biarticular flexibility (knee extended) 86.6° ± 8.6°. Monoarticular hip flexibility (knee flexed) 127.8° ± 12°.	TAS score 4.7 ± 0.8.	NR	Sport types: preinjury 3.3 ± 1.4, follow-up 3.0 ± 0.7. Training frequency (sessions/wk): preinjury 2.0 ± 1.4, follow-up 2.6 ± 0.7. Training duration (min/wk): preinjury 154 ± 83, follow-up 129 ± 64.	VAS score 0.7 ± 0.9	NR
Chochołaś ⁹	NR	Median EQ-5D-5L score 0.94 (range, 0.83-1.00); LEFS score 88.8 (range, 61.6-100.0); PHAT overall score 78.8 (range, 54.6-99.8); LSI of flexor strength 95.6% (range, 71.7%-135.6%).	Median maximum flexor muscle strength 0.82 Nm/kg (range, 0.33-1.42) on the operated side vs 0.72 Nm/kg (range, 0.41-1.89) contralaterally (<i>P</i> = .807). Median LSI 95.6% (range, 71.7%-135.6%), corresponding to a 3.5% strength deficit.	Median passive hip flexion (knee flexed) 120° (1 range, 15°-140°) operated vs 120° (range, 115°-150°) contralateral (<i>P</i> = .581). Median passive hip flexion (knee extended) 90° (range, 60°-120°) operated vs 90° (range, 70°-110°) contralateral (<i>P</i> > .999).	NR	All patients were very to extremely satisfied with surgery, with a median satisfaction rate of 100% (range, 90%-100%).	NR	VAS sitting score median 1.0 (range, 0.0-8.1); VAS rest score median 0.0 (range, 0.0-4.2).	1 superficial wound infection treated surgically; 1 local skin dehiscence excised and resutured; no tendon reruptures or sciatic radiculopathy at follow-up.
Chahal ⁸	NR	Postoperative mean LEFS score 74.9 ± 7.8 (range, 59-80); 53.8% scored 80/80. Postoperative HHS 90.7 ± 13.9 (range, 67-100); 8 excellent, 1 good, 4 poor (all female, 1 repaired after 4 y).	Isokinetic muscle testing in 12 patients showed 78.0% ± 6.1% recovery (range, 74.0%-88.0%) vs contralateral side; extension deficit trend was nonsignificant (<i>F</i> = 1.14; <i>P</i> = .35).	All patients had symmetrical walking (12/12). Calf circumference 57.5 cm operated vs 57.2 cm contralateral. Mid thigh circumference 58.5 cm operated vs 59 cm contralateral (NS).	ROM measured with a goniometer showed no significant difference between injured and uninjured leg.	All patients (13/13) were extremely satisfied with the surgery.	SANE recovery score: 93.6% ± 7.5% (range, 75.0%-100.0%).	VAS score 1.3 ± 1.9 (range, 0-5); 3 (23%) reported pain in past week; 1 (8%) reported painless knot, 1 (8%) reported discomfort after sitting >45 min, 1 (8%) used pain medication 1-2 times/wk, and 1 (8%) rarely used medication (few days/mo).	3 (23%) reported constant hamstring stiffness; 3 (23%) reported morning stiffness; 1 reported mild numbness/tingling below knee (not bothersome).
Fouasson-Chailloux ¹²	NR	RTS at 7.0% ± 2.9 mo (same or lower level). Mean LEFS score >71 in 14 patients; LEFS score 76.6 ± 3.9. Mean UCLA score 37.5 ± 2.2. Activity level 8.7 ± 1.4. Residual pain 8.7 ± 1.3.	Concentric strength deficit was 40.0% ± 25% at 4 mo, improving to 25.0% ± 12% at 2 y; eccentric deficit persisted.	NR	TAS score decreased from 6.9 ± 1.7 to 6.1 ± 1.9 (<i>P</i> = .03).	15 patients were very satisfied or satisfied with high subjective scores.	7.0 ± 2.9 mo (range, 4.0-12.0 mo).	NA	14 patients had ecchymosis.
Ayuob ¹	NR	Surgical intervention improved absolute PSLR and reduced PSLR deficit at 3 mo. At 3 mo, 15% had LEFS score 80/80, 20% >75. At 2 y, 30% had LEFS score 80/80, 45% >75. MARS scores improved significantly at all follow-ups.	At 1 y, all patients restored >90.0% strength of the contralateral side.	PSLR was 71.5° ± 6.5° at 3 mo and 91.5° ± 9.3° at 1 y. PSLR deficit improved from 25.0° ± 11.4° at 3 mo to 5.0° ± 8.3° at 1 y. 2 gymnasts and 1 dancer had a residual 20° deficit at 1 y but returned to sport after additional rehabilitation.	NR	19 (95%) patients reported being very satisfied or satisfied at follow-up.	19/20 (95.0%) returned to preinjury sporting levels. Return time 11.9 ± 5.7 wk. No recurrence reported.	NR	4 (20%) postoperative hematomas diagnosed clinically; 1 superficial wound infection treated with 1 wk of oral antibiotics.

(continued)

TABLE 3
(continued)

Study	Groups	Functional Recovery	Muscle Strength	ROM	Functional Activities	Patient Satisfaction	Return to Preinjury Activity Levels	Pain Levels	Adverse Effects or Complications
Ayuob ²	NR	LEFS and MARS scores improved at 3 mo, 1 y, and 2 y. At 1 y, 10.9% had LEFS score 80/80, 51.6% >75. At 2 y, 31.3% had LEFS score 80/80, 57.8% >75. At 3 mo, 5 patients had PSLR <30° (2 MTJ-BFh myofascial tears, 1 complete rerupture, 2 chronic lumbar pain). At 1 y, only 1 patient (chronic lumbar pain) did not achieve PSLR >30°. PSLR improved further at 1 y vs 3 mo.	Surgical intervention improved strength at 3 mo. By 1 y, all restored >90.0% strength.	PSLR was 72.0° ± 11.4° at 3 mo and 77.8° ± 7.2° at 1 y. Deficit decreased from 11.5° ± 9.8° to 5.0° ± 8.3°. At 3 mo, 5 patients had PSLR <30° (2 myofascial MTJ-BFh tears, 1 rerupture, 2 chronic lumbar pain). At 1 y, only 1 patient with chronic lumbar pain did not achieve PSLR >30°. Further PSLR improvement was observed at 1 y vs 3 mo.	NR	Surgical repair of MTJ-BFh injuries yielded high satisfaction at 1 and 2 y. At 1 y, 2 patients were unsatisfied (1 rerupture requiring reoperation, 1 slow rehabilitation). At 2 y, 81.2% very satisfied and 18.8% satisfied.	RTS 13.4 ± 5.1 wk. At 1- and 2-y follow-ups, all patients still participated at preinjury levels.	NR	3 patients with recurrence; (4.7%); 12 had postoperative hematomas distal to incision; 4 had posterior cutaneous nerve neuraupraxia. All resolved within 8 wk.
Mansour ²²	NR	NR	All 10 players regained 5/5 strength bilaterally; 3 had symmetrical isokinetic results. 9/10 returned to play, although 1 reported loss of speed and could not compete. 4 played only 1 game (reasons included backup role, unrelated injuries, or being cut from team).	NR	NR	NR	9/10 players returned to play next season, but only 5/10 played >1 game despite no surgical limitations.	NR	NR
Johnson ¹⁹	Workers' compensation patients	Mean ± SD LEFS score 80.3 ± 20.2; custom LEFS score 66.9 ± 25.9.	80.6 ± 20.1	NR	MARS score 3.5 ± 4.3; custom MARS score 88.7 ± 29.7.	93.1 ± 13.1	NR	VAS score 3.5 ± 3.7	NR
	Control	Mean ± SD LEFS score 94.3 ± 10.1; custom LEFS score 88.6 ± 12.0.	91.8 ± 10.6	NR	MARS score 4.4 ± 4.4; custom MARS score 95.7 ± 12.1.	96.7 ± 5.3	NR	VAS score 4.0 ± 4.0	NR
Bowman ⁵	NR	iHOT-12 score 99; KJOC score 77; median SANE score 90. 78% reported good/excellent results. SANE ADL score improved by 29 points; SANE activity score by 39 points.	NR	NR	TAS scores: 5.5 preoperatively, 5.1 postoperatively.	Overall satisfaction was 94.0%.	At 6.3 mo, 82.0% of runners returned to running. At 7.6 mo, 88.0% resumed usual sports, 72.0% (42/58) at same level.	VAS score 1.6 ± 2.2	Overall complication rate 15.0%. A total number of 9 complications were identified: 1 major (wound dehiscence requiring debridement) and 8 minor (3 infections, 2 persistent numbness, 4 ongoing cramping/pain/fatigue).
Lefevre ²¹	NR	UCLA score 40; preinjury 9.7 ± 1.2, preoperatively 3.9 ± 1.9, postoperatively 9.7 ± 0.8.	NR	NR	TAS scores: 8.9 ± 1.9 preinjury, 2.9 ± 2.1 preoperatively, 8.5 ± 2.4 postoperatively.	NR	RTS at 1 y: yes 63 (98.4%), no 1 (1.6%). Performance: same 50 (78.1%), inferior 12 (18.8%), stopped 2 (3.1%).	NR	NR
Subbu ²¹	Early Intervention (<6 weeks)	NR	NR	NR	NR	NR	Athletes returned to play: 77/78 (98.7%). Mean return: 16 wk (range, 12-32 wk).	NR	Complications: 4 superficial wound infections, 2 local neural symptoms.

^aADL, activities of daily living; BMI, body mass index; HHS, Harris Hip Score; iHOT-12, International Hip Outcome Tool-12; KJOC, Kerlan-Jobe Orthopaedic Clinic; LEFS, Lower Extremity Functional Scale; LSI, limb symmetry index; MARS, Marx Activity Rating Scale; MTJ-BFh, musculotendinous junction of the long head of the biceps femoris; NR, not reported; NS, not significant; PHAT, Perth Hamstring Assessment Tool; PSLR, passive straight-leg raise; ROM, range of motion; RTS, return to sport; SF-12, Short Form-12; SANE, Single Assessment Numeric Evaluation; TAS, Tegner Activity Scale; UCLA, University of California, Los Angeles; VAS, visual analog scale; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index.

TABLE 4
NOS Quality Assessment Results^a

Study	Selection	Comparability	Outcome	Total Score
Shambaugh ³⁰	★★★		★★★	6 (good quality)
Johnson ¹⁹	★★★	★★	★★	7 (good quality)

^aNOS, Newcastle-Ottawa Scale.

Return to Preinjury Activity Levels

Operative Treatment. Chahal et al⁸ found a mean Single Assessment Numeric Evaluation score of $93.6\% \pm 7.5\%$ (range, 75.0%-100.0%), although only 16 of 27 (59.3%) patients fully returned to preinjury activity. In 1 study, 90.0% (9/10) resumed play despite full strength recovery in all cases.²² A return-to-sport (RTS) rates of 98.7% (77 patients) was reported in another study,³¹ and 14 of 15 patients resumed their preinjury level in the study by Sandmann et al.²⁹ In their work, Willinger et al³³ found that RTS outcomes varied, with 50.0% returning to the same level, 41.7% returning to a reduced level, and 8.3% ceasing participation in sport. Bowman et al⁵ reported an RTS rate of 88.0% at a mean of 7.6 months, with 72.0% (42/58) resuming at their prior level. In their work, Ayuob et al^{1,2} observed RTS rates of 95.0% (19/20) at a mean of 11.9 ± 5.7 weeks and 100.0% at a mean of 13.4 ± 5.1 weeks. In the study by Lefèvre et al,²¹ 63 (98.4%) patients returned to sport, with 78.1% maintaining and 18.8% reducing their performance level.

Operative Versus Nonoperative Treatment. All surgically treated patients returned to their preinjury activities, compared with 8 of 11 in the nonoperative group.³⁰

Pain Levels

Among the 7 articles that addressed pain levels, the lowest mean VAS score was 0.7 ± 0.9 ,²⁹ whereas the highest was 4.0 ± 4.0 after surgical intervention.¹⁹

Adverse Effects or Complications

Operative Treatment. In 1 study, 6 cases of postoperative stiffness and 1 case of numbness/tingling were reported.⁸ In the study by Subbu et al,³¹ 6 complications were noted, including 4 superficial infections and 2 neural symptoms. In another study, 3 (4.2%) complications and sitting discomfort in 36.6% of patients were documented.³³ A 15.0% complication rate was reported by Bowman et al,⁵ including 1 major complication (wound dehiscence) and 8 minor complications. In the work by Ayuob et al,¹ 4 (20.0%) patients developed hematomas, and 1 had a superficial infection. The same group also observed 12 hematomas, 3 recurrences, and 4 cases of neurapraxia.² Two additional postoperative complications were reported in the work of Chocholáč et al.⁹

Nonoperative Treatment. Ecchymosis was observed in 14 patients managed nonoperatively.¹²

Operative Versus Nonoperative Treatment. In the study by Shambaugh et al,³⁰ 2 patients in the operative group developed postoperative complications. One patient experienced a hematoma, which required a reoperation, while the other patient developed incisional numbness, which resolved within 3 months after surgery.³⁰

Risk of Bias

Cohort Studies. Quality assessment indicated that 2 studies^{19,30} were of overall good quality. The study by Johnson et al¹⁹ achieved the highest score (7/9), reflecting a strong methodology across the selection, comparability, and outcome domains. The study by Shambaugh et al³⁰ scored 6 of 9, with adequate performance in selection and outcome assessment, but limited comparability (Table 4).

Case Series Studies. Eleven case series were assessed using a 9-item standardized checklist. Nine studies^{1,2,5,8,9,12,21,31,33} each scored 7 of 9, demonstrating strong reporting across inclusion criteria, valid measurements, and follow-up data. One study²⁹ scored 6 of 9, with limitations in inclusion clarity and site data. The study by Mansour et al²² had the lowest score (5/9), showing deficiencies in inclusion criteria and demographic reporting (Table 5).

DISCUSSION

The major findings of our study demonstrated that proximal hamstring avulsions are significant lower limb injuries leading to impaired strength, reduced function, and limitations in both daily and athletic activities.²⁴ We found that the 2 primary management approaches for these injuries are operative and nonoperative treatment. Surgical repair typically involves reattaching the avulsed tendon to the ischial tuberosity, while nonoperative treatment emphasizes physical therapy and gradual return to function.^{24,26} Despite the availability of both approaches in clinical practice, their comparative effectiveness remains an area of ongoing investigation and debate.

The major findings of our review demonstrated that surgical intervention was frequently associated with favorable outcomes in terms of functional recovery, strength, and return to activity. Across several validated scoring systems, including the LEFS score, HHS, Lysholm score, WOMAC, and UCLA score, patients in the surgical groups generally reported higher scores and greater recovery, although direct comparisons with nonoperative management were limited in several studies.^{1,2,5,8,9,15,19,21,29} These trends were observed in both short-term and long-term follow-ups,

TABLE 5
Case Series Checklist Quality Assessment Results

Study	Were There Clear Criteria for Inclusion in the Case Series?	Was the Condition Measured in a Standard, Reliable Way for All Participants Included in the Case Series?	Were Valid Methods Used for Identification of the Condition for All Participants Included in the Case Series?	Did the Case Series Have Consecutive Inclusion of Participants?	Did the Case Series Have Complete Inclusion of Participants?	Was There Clear Reporting of the Demographics of the Participants in the Study?	Was There Clear Reporting of Clinical Information of the Participants?	Were the Outcomes or Follow up Results of Cases Clearly Reported?	Was There Clear Reporting of the Site(s)/Clinic(s) Demographic Information?	Score
Willinger ³³	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Unclear	7
Sandmann ²⁹	Unclear	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Unclear	6
Chocholáč ⁹	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Unclear	7
Chahal ⁸	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Unclear	7
Fouasson-Chailloux ¹²	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Unclear	7
Ayuob ¹	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Unclear	7
Ayuob ²	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Unclear	7
Mansour ²²	Unclear	Yes	Yes	Unclear	Yes	Unclear	Yes	Yes	Unclear	5
Bowman ⁵	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Unclear	7
Lefevre ²¹	Unclear	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Yes	7
Subbu ³¹	Unclear	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Yes	7

particularly in studies reporting LEFS and MARS scores,^{1,2} suggesting sustained postoperative function. However, due to incomplete reporting of comparative nonoperative outcomes in many studies, definitive conclusions regarding the superiority of surgical management could not be drawn.

In contrast, nonoperative treatment produced moderate improvements. While LEFS scores approached those of surgical cohorts, overall functional gains were less pronounced, and activity-specific outcomes, such as hop performance and UCLA scores, remained inferior.¹² Better LEFS and strength values were observed in surgically treated patients.³⁰ It is important to note that these comparisons rely on a significantly unequal distribution of data—11 operative studies compared with only 1 nonoperative cohort—thereby limiting the validity and generalizability of direct comparisons. These differences reflect a recurring pattern in the literature, where nonoperative management yields acceptable but suboptimal results, particularly in high-demand populations.^{3,18}

Muscle strength recovery was also superior in the surgical group. Most studies reported that patients regained $\geq 75.0\%$ to 90.0% of contralateral leg strength, primarily when surgery was performed acutely.^{1,2,8,15,19} Balanced flexor strength and high LSI values were also reported, further supporting the role of surgical repair in restoring muscle function.⁹ This trend is reinforced by findings of significantly higher strength outcomes in operative patients (85.0%) compared with nonoperative ones (64.0%).³

In contrast, nonoperative cohorts experienced persistent strength deficits, particularly in eccentric and isometric contraction.¹² Delayed or nonoperative management was associated with poorer strength recovery.¹⁵ Furthermore, heterogeneity in strength testing protocols (eg, isokinetic vs isometric) across studies limits direct comparison.³⁰

Regarding ROM, surgical patients generally maintained near-normal joint mobility, with no significant differences between the injured and uninjured sides. This indicates that early surgical repair does not appear to compromise joint flexibility and may facilitate an early return to activity.^{9,15,29} Comparable passive hip flexion was reported on both operated and nonoperated sides ($P = .581$ and $P > .999$), reinforcing the preservation of ROM after repair.⁹ However, no data were available for ROM after nonoperative treatment, highlighting a notable gap in the literature. Future studies should address whether nonoperative treatment affects passive or active ROM.

Postoperative functional activity levels were generally favorable after surgical repair. Most studies reported a return to moderate to high activity levels, with TAS scores ranging from 4.7 to 6.^{5,29,33} Higher performance was also observed, with a mean TAS score of 8.5 ± 2.4 .²¹

In contrast, nonoperative treatment resulted in a statistically significant decline in activity level.¹² While comparable SF-12 mental and physical scores were reported, objective performance and RTS rates favored the operative group.³⁰ These inconsistencies between subjective and objective measures underscore the need for standardized activity assessments.

Complications were more frequent in surgical cases but were largely minor, including transient stiffness, hematomas, or superficial infections.^{1,2,5,8,31} Lower complication rates were observed in acute compared with chronic repairs.^{15,18} A complication rate of 23.2% was reported among surgically treated patients, underscoring the importance of careful patient selection.³ Overall, the safety profile is considered acceptable when weighed against functional benefits. Although nonoperative management is less invasive, complications such as prolonged bruising or soft tissue issues were reported.¹² However, underreporting remains a possibility. Longitudinal studies may

better capture delayed complications, including chronic weakness or reinjury.

Implications for Clinical Practice and Future Research

The findings of this systematic review suggest that operative treatment is associated with better functional, strength, and RTS outcomes compared with nonoperative management, especially in active individuals or athletes. However, treatment decisions should be personalized, taking into account patient goals, comorbidities, and risk profiles.

To improve clinical decision-making and evidence quality, future research should prioritize prospective comparative studies using standardized outcome measures, including consistent assessment of ROM, strength, and functional activity. Moreover, further investigation into the long-term effects and efficacy of nonoperative management is essential to clarify its role and identify patient populations most likely to benefit from nonoperative approaches.

Limitations

This review has several limitations that should be acknowledged. Many included studies had small sample sizes and lacked control groups, reducing external validity. There was substantial heterogeneity in outcome measures, follow-up durations, and strength assessment protocols, making meta-analysis unfeasible. Most of the included studies were observational, and only English-language articles were considered, introducing language bias. It is also important to note that these comparisons rely on a significantly unequal distribution of data—11 operative studies compared with only 1 nonoperative cohort—thereby limiting the validity and generalizability of direct comparisons. Furthermore, the specific tear patterns in each group were not clearly reported; therefore, it is unclear whether the operative series included only complete proximal tendon tears or whether muscle injuries were also represented. This limits the ability to make direct comparisons between operative and nonoperative cohorts. The relatively small number of nonoperative cases, particularly for muscle-related injuries, further reduces the strength of any conclusions drawn and highlights the need for caution in interpretation, including consideration of whether the title should more accurately reflect the available evidence base. Additionally, the exclusion of gray literature, unpublished data, and studies published before 2000 may have introduced publication bias. Taken together, these factors indicate that strong conclusions are difficult to make, and the findings of this review should be interpreted with appropriate caution.

CONCLUSION

Our study demonstrated that operative treatment resulted in generally good functional outcomes, substantial muscle strength recovery, high patient satisfaction, and a high rate of RTS, although with a higher incidence of mostly minor complications that were not statistically significant.

Nonoperative management also showed good outcomes in the few studies available. Given the limited quantity and heterogeneity of available studies, particularly for nonoperative treatment, the current literature does not allow definitive recommendations favoring one approach over the other. Prospective studies with standardized outcome measures and clear subtype definitions are needed to clarify which patients derive the most benefit from operative versus nonoperative treatment.

Artificial Intelligence (AI) Use

AI was used to assist with language correction, specifically grammar and improving the clarity of writing, as English is not our native language. The program utilized was ChatGPT (Version GPT-5.1) [Large language model]. <https://www.openai.com/>.

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APPENDIX 1.

The following Boolean search strings and MeSH terms were used across the included databases:

- PubMed: (“Hamstring muscle injury” OR “Hamstring rupture”) AND (“Surgical treatment” OR “Conservative treatment”).
- Web of Science (Core Collection): TS=(“hamstring muscle injury” OR “hamstring muscle rupture” OR “hamstring injury”) AND TS=(“operative treatment” OR “conservative treatment”) AND LA=(English).
- Google Scholar: (“Hamstring muscle injury” OR “Hamstring muscle rupture” OR “Hamstring injury”) AND (“Operative treatment” OR “Conservative treatment”).
- ScienceDirect: (“Hamstring Muscle Injury” OR “Hamstring Rupture” OR “Hamstring Injury”) AND (“Surgical Treatment” OR “Conservative Treatment”).